

Tanmay Vij

Lead Software Engineer | Systems Architect: AI-Native & Distributed Systems

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PROFESSIONAL SUMMARY

Lead Software Engineer and Systems Architect with 6+ years building production AI and distributed systems, including 5 years of technical leadership with end-to-end architectural ownership. Designs AI-native platforms - RAG pipelines, vector search, agentic workflows, LLM orchestration, and production inference, alongside resilient multi-tenant cloud architectures serving multi-billion-dollar enterprises. Delivered measurable outcomes: cut AWS costs by up to 70%, modernized mission-critical legacy platforms, and scaled a startup to 1,000+ monthly users within 6 weeks of launch. AWS Certified Solutions Architect (Associate).

WORK EXPERIENCE

Lead Software Engineer - Platform Architecture & Delivery

May 2021 - Present

Softobotics Technologies | Bangalore, India

Lead technical owner across multiple enterprise SaaS engagements — accountable for architecture decisions, delivery roadmaps, client relationships, hiring-level mentorship, and production operations. Primary stack: .NET Core, Java (Spring Boot), Node.js (NestJS), AWS.

- **Cut monthly AWS spend by up to 70%** for a resource-constrained client by refactoring system architecture, redesigning application delivery, and aligning cloud resources with actual workload characteristics, while improving latency and preserving security, scalability, and high availability.

RecordsKeeper.ai - multi-tenant AI-native RAG SaaS

- Designed and built a multi-tenant RAG platform with automated document ingestion, embedding pipelines, vector search, and continuous indexing, enabling each organization to maintain its own isolated, continuously updated knowledge base from incoming documents.
- **Reduced document discovery from hours of manual search to seconds** via conversational retrieval over sensitive document collections, while enforcing strict tenant isolation, SOC 2 and ISO 27001 compliance requirements and data confidentiality.

Opus Bilprovning - Swedish vehicle inspection platform

- Executed end-to-end modernization of a legacy monolithic .NET Framework estate into modular .NET Core REST APIs and worker services, delivering a scalable, cloud-ready backend serving a nationwide network of inspection stations with an Angular frontend.
- Architected distributed data-processing workflows and rebuilt critical execution paths around idempotency, parallelism, and reliable event delivery; optimized complex SQL Server queries, cutting latency across high-throughput inspection and reporting pipelines.
- Migrated SQL Server from database mirroring to Always-On Availability Groups, materially improving availability and fault tolerance for mission-critical workloads.
- Led production incident response and root-cause analysis with minimal downtime; partnered with DevOps to redesign environment architecture and deployment workflows, improving release reliability and operational efficiency.
- Served as primary technical point of contact for the client, owning requirements, sprint planning, and mentorship while aligning engineering output with business priorities.

Repositree - multi-tenant commercial real estate SaaS

- Re-architected a tightly coupled hybrid system (monolith plus fragmented microservices) into a modular service-based backend with clear service boundaries, centralized IAM and billing, CI/CD, and observability - **supporting production traffic for multi-billion-dollar real-estate firms**.
- Designed claim-based multi-tenant architecture with strict data isolation, access control, and tenant-aware workflows with clean separation of concerns across services.
- Optimized database schemas and queries across a distributed system, and leveraged caching to reduce redundancy and improving reliability for multiple enterprise tenants.

Catering Rewards - *high-volume restaurant loyalty platform*

- Built containerized microservices on AWS (SQS, Cloud Map, Terraform) supporting asynchronous, high-throughput order processing and real-time payment processing.
- Implemented tenant-aware authentication and role-based access control (RBAC), ensuring secure data isolation across partners; designed transactional workflows including a double-entry ledger system, guaranteeing ACID-compliant financial data integrity and automated reconciliation.

Foundercrate - *startup fundraising and investor CRM platform*

- Built an investor CRM with Kanban workflows from scratch, plus a bulk emailing engine on AWS SQS and SES with throttle-aware queue processing to operate within SES daily send limits.
- Optimized investor database queries over nested tables exceeding 200K records; engineered nested AWS CloudFormation stacks to work around the 500-resource hard limit.

Co-Founder & Director

Oct 2020 - May 2021

RePrice Technologies | India

- Designed the full system architecture and led development and deployment of backend APIs and Android/iOS mobile apps (.NET Core, Firebase, React Native); **scaled the platform to 1,000+ users/month within 6 weeks of launch.**
- Integrated third-party pricing APIs across multiple service providers; owned product strategy, customer feedback loops, and business operations end to end.

Full Stack Engineer & Cloud Architect

Apr 2020 - Apr 2021

Independent Consultant | Remote

- Designed real-time chat and meeting modules (WebSockets, WebRTC) for a Next.js productivity platform engineered for low-bandwidth reliability; built Node.js backends supporting thousands of concurrent users with minimal latency.
- Automated VM provisioning and configuration on DigitalOcean with infrastructure-as-code templates, enabling seamless tenant onboarding.
- Built a WebSocket-based research survey chat platform with dynamic participant assignment, chatbot and Qualtrics integrations, and analytics pipelines for engagement tracking and data export.

SKILLS

Languages: C#, Python, TypeScript, JavaScript, SQL

AI & LLM Systems: Retrieval-Augmented Generation (RAG), Vector Search & Vector Databases, Embedding Pipelines, Agentic Workflows, LLM & Model Orchestration, Inference Optimization, Fine-Tuning

Frameworks: .NET 8, ASP.NET Core, FastAPI, Node.js, NestJS, React, Angular

Architecture: Microservices, Event-Driven Architecture, Multi-Tenant SaaS, Distributed Systems, Serverless, REST & GraphQL API Design, Domain-Driven Design

Messaging: Apache Kafka, RabbitMQ, AWS SQS/SES, Pub/Sub Messaging

Databases: SQL Server, MySQL, MongoDB, Redis, Elasticsearch

Cloud & Infrastructure: AWS (Lambda, SQS, SES, CloudFormation, ECS, Cloud Map), Azure, GCP, Docker, Kubernetes, Terraform, Infrastructure as Code, CI/CD

Practices: High Availability, Security & Compliance, Monitoring & Observability, Incident Response, Cost Optimization, Agile/Sprint Leadership

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate, Amazon Web Services (Aug 2025)
- Deep Learning Certification — One Fourth Labs, in collaboration with IIT Madras (Mar 2020)

EDUCATION

B.Tech in Computer Science and Engineering

Amity University, Noida, India

LANGUAGES SPOKEN

English, French, Spanish